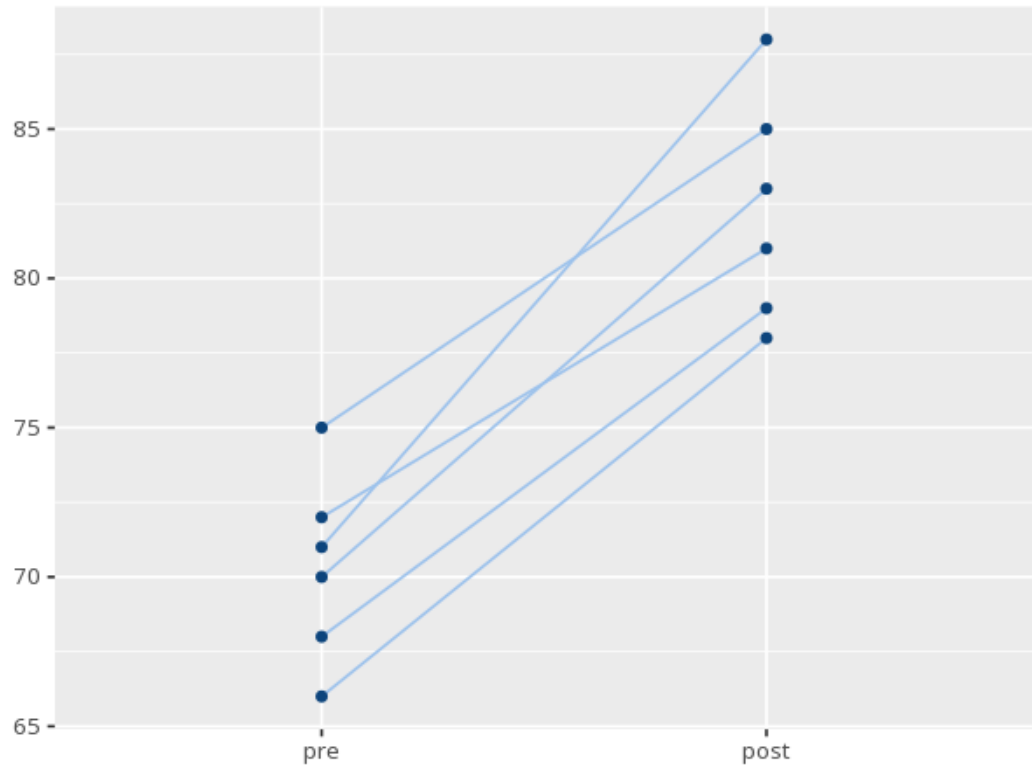


1 Paired t-test

Condition	N	M	SD
pre	6	70.33	3.14
post	6	82.33	3.78

Pair	t	df	p	Mdiff	95% CI	d [95% CI]z	r (paired)
pre – post	–10.39	5	<.001	–12.00	[–14.97, –9.03]	–4.24 [–6.90, –1.58]	0.68



Tests whether the average change between two related measurements (e.g. before vs. after) differs from zero. A p below α means a significant change; the mean difference, CI and d_z describe its size (their sign depends on the subtraction order, $A-B$). A p at or above α does not prove there was no change — only that none was detected. Your significance threshold (α) is 0.05.

A paired-samples t -test gave $M=-12.0$, $t(5)=-10.39$, $p < .001$, $d_z=-4.24$ [–6.90, –1.58].

Statistical basis: Paired-samples t -test (Student (1908). "The probable error of a mean." *Biometrika*, 6(1), 1-25.); Cohen's d_z effect size (Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Routledge.); Paired correlation (Pearson r) between the two conditions (Pearson, K. (1895). "Note on regression and inheritance in the case of two parents." *Proceedings of*

the Royal Society of London, 58, 240-242.).